

Beyond Memorization: A Neuro-Symbolic Strategy for Verifiable Chain-of-Thought Reasoning in Symbolic Puzzles

Deconstructing the Baseline: The Role of Memorization vs. Genuine Reasoning

The primary obstacle to improving performance on the specified puzzle dataset is not a simple deficiency in the model's reasoning capabilities but a deeply ingrained reliance on a pattern-matching mechanism that has been optimized to achieve a 0.86 leaderboard score . This baseline represents a local optimum built on memorization rather than symbolic reasoning, a critical distinction that fundamentally reframes the research goal from "improving CoT" to "replacing a flawed reasoning paradigm." The discovery that the baseline CoTs are "fake"—specifically, all-zeros mappings—reveals that the model has learned to associate puzzle inputs directly with correct answers without generating any intermediate logical steps . This finding is consistent with established research indicating that Large Language Models (LLMs) often exhibit a competitive dynamic between deliberate, step-by-step reasoning (as elicited by CoT prompting) and direct retrieval from internal memory ³ . Models fine-tuned with distillation techniques are particularly susceptible to this behavior, often resorting to "post-hoc explanation," where they fabricate plausible-sounding rationales after having already arrived at a memorized answer ³ . The current baseline appears to have perfected this shortcut, achieving high accuracy through pattern recognition alone.

This memorization-based approach creates a fragile equilibrium that is highly sensitive to perturbation. Any attempt to introduce more authentic CoTs disrupts this finely tuned mechanism, leading to a regression in overall performance. For instance, the experiment that replaced 428 cryptarithm CoTs resulted in a drop in the LB score to 0.83 . This outcome strongly suggests that the original, "fake" CoTs were serving a crucial function for those specific instances, likely acting as a key to unlock the memorized answer. Replacing them with real CoTs broke this link, causing the model to fail on those puzzles while perhaps not yet being able to solve them correctly through genuine reasoning. This phenomenon underscores the danger of treating the problem as a simple matter of data augmentation. Simply adding more examples with real CoTs is insufficient because it

risks overwhelming the model's existing, successful-but-flawed strategy. The failure of the 78K mega-dataset strategy, which diluted the signal and led to a score of 0.76 due to "format chaos," further illustrates this point. The model's success is not distributed across many different reasoning styles; it is concentrated in a narrow, format-dependent pattern of memorization.

The concept of "contextual drag" provides a theoretical framework for understanding why mixing real and fake CoTs is so detrimental [21](#). When a model is presented with inconsistent or misleading information within its context window, its ability to perform accurate reasoning can degrade significantly. In this case, the "fake" CoTs act as a form of noise or contextual drag for the puzzles where real CoTs are being introduced. The model's internal processing pipeline seems to depend heavily on the specific formatting and structure of the input-output pairs it was trained on. Deviating from these learned formats, even if the logical content of the new CoT is correct, can break the execution flow and prevent the model from arriving at the right answer. This highlights the brittleness of the current system and points toward a need for a much stricter, more structured CoT format. Research into Structured CoTs (SCoTs) for code generation demonstrates the power of imposing a formal grammar or template on reasoning traces to guide the model's output and improve reliability [13](#). Similarly, benchmarks designed for verifiable reasoning, such as FinChain, emphasize the importance of step-level consistency alongside final-answer correctness [31](#) [50](#). The current baseline, by relying on a uniform "all-zeros" mapping, inadvertently created a format that the model could exploit for memorization. Any new strategy must therefore prioritize format consistency above all else to avoid reintroducing the chaos that degraded performance in previous attempts.

Diagnosing Failure Modes in Proposed CoT Strategies

The systematic testing of 15 hypotheses provides a rich diagnostic dataset that reveals three primary failure modes preventing the improvement of the baseline: Format Inconsistency, Loss of Memorization, and Structural Mismatch. These categories encapsulate the core challenges of modifying a system that functions primarily through memorization rather than generalizable reasoning. Understanding these failure modes is essential for designing a successful intervention.

Format Inconsistency is a recurring and critical failure mode. Multiple strategies that attempted to introduce real CoTs failed because they violated the specific formatting

conventions implicitly learned by the model. For example, RFT Round 1, which used model-generated self-CoTs, resulted in a lower LB score of ~ 0.82 - 0.85 specifically because the generated CoTs had a "Wrong format vs baseline". Similarly, the integration of solvers for bit-manipulation GF(2) problems (Strategy 9) and plain-digit ciphers (Strategy 10) both introduced format inconsistencies that caused partial regressions and failed to yield an overall improvement. This indicates that the model's solution process is not purely semantic; it is tightly coupled with syntactic and structural cues present in the training data. The model may have learned that certain tokens or formatting patterns serve as markers to trigger its memorization-based lookup table. Disrupting this format, even with logically sound reasoning, effectively severs the connection between the puzzle and the stored answer. This aligns with findings that LLMs can develop brittle dependencies on superficial features of their training data, leading to poor generalization when those features change [6](#).

Loss of Memorization is another major failure category, directly linked to the baseline's reliance on pattern matching. The most telling example is the v2-replace 428 cryptarithm CoTs experiment, which dropped the LB score to 0.83 . This demonstrates that the original "fake" CoTs were instrumental for solving those particular puzzles. Replacing them destroyed the memorization signal, and since the model was not simultaneously equipped with the correct reasoning capability to solve them anew, it resulted in a net loss. This principle explains the catastrophic failure of the 78K mega-dataset strategy (Strategy 5), which, despite its size, suffered from "too diverse styles, format chaos" and saw the score plummet to 0.76 . The sheer diversity of formats and styles overwhelmed the model's capacity to maintain its specialized memorization patterns, diluting the signal for the majority of the dataset. This reinforces the idea that the current 0.86 score is a hybrid of true reasoning on a few puzzle types and pure memorization on the vast majority. A naive approach of indiscriminately replacing fake CoTs with real ones risks collapsing the entire memorization component, leading to a significant performance drop unless the new reasoning module is immediately perfect and universally applicable.

Structural Mismatch and a lack of generalization represent a third, more subtle failure mode. Even when strategies were structurally sound and avoided format issues, they often failed to transfer knowledge or learn new patterns. For instance, creating a new `equation_symbolic` category (Strategy 7) and deploying long-form hypothesis-testing CoTs (Strategy 8) had no discernible effect on the LB score, suggesting the model did not learn the new pattern or generalize the new reasoning style. This points to a deeper issue: the model struggles to integrate novel reasoning paradigms into its existing cognitive framework. It may be capable of generating fluent text that resembles a CoT but lacks the underlying logical structure required for symbolic manipulation. This is a classic problem of faithfulness in CoT, where the generated trace does not accurately

reflect the model's actual decision-making process 34 52 . The failure of the GRPO RL attempt, which yielded no variance and 0/4 exact rewards, further supports this conclusion . The reward signal was likely too sparse or poorly defined to guide the model towards a more robust reasoning strategy. This mirrors findings from benchmarks like EngChain, where top models exhibited a significant "right for the wrong reasons" gap, demonstrating high textual fluency but critically low procedural reasoning capability 44 . The dominant error type was conceptual, not computational, indicating that models often fail by misapplying principles rather than making simple mistakes 44 .

Hypothesis	Dataset Size Change	LB Result	Primary Failure Mode
Tong-style baseline	No change	0.86 ✓	Memorization + Format Consistency (Local Optimum)
RFT Round 1 (self-CoTs)	+703	~0.82-0.85	Format Inconsistency
Donald-format generators	+5,500	0.80	Format Mixing / Chaos
78K mega-dataset	+67,745	0.76	Loss of Memorization & Format Chaos
New equation_symbolic cat.	+839	0.86	Lack of Generalization
Long-form hypothesis-testing CoTs	+839	0.86	Lack of Generalization
Bit-manipulation GF(2) solver	+987	0.85	Format Inconsistency
v2-replace 428 cryptarithm CoTs	-395	0.83	Loss of Memorization
All-or-nothing real CoTs	-395	Not Yet Tested	High-risk structural replacement

Synthesizing a Neuro-Symbolic Approach for Verifiable Reasoning

To overcome the limitations of the current baseline and the failure modes of past interventions, a strategic shift away from heuristic data augmentation is necessary. The path forward lies in adopting a neuro-symbolic approach that prioritizes verifiable, structured reasoning over unstructured language generation. This methodology involves moving beyond generating plausible-sounding CoT traces to producing machine-checkable, step-by-step proofs for each puzzle. Such an approach is grounded in several key principles from recent research: the use of programmatic solvers for ground-truth generation, the enforcement of strict structural consistency, and the implementation of a verification loop to ensure faithfulness.

The first pillar of this new approach is the development of ground-truth solvers for the failing puzzle types. Instead of relying on potentially ambiguous human-generated CoTs,

the strategy should involve creating deterministic, programmatic engines for each puzzle category, including cipher, bit-manipulation, and equation_symbolic. These solvers would be responsible for not only finding the correct answer but also generating a perfectly consistent and verifiable chain of reasoning that leads to it. This technique is central to frameworks like rStar-Math, which uses Monte Carlo Tree Search (MCTS) with a policy model and a process reward model to generate millions of high-quality, step-by-step solutions for math problems [47](#). By executing these solutions as code, the system can automatically detect computational and symbolic deduction errors, ensuring the highest quality of training data [47](#). Similarly, a hybrid framework integrating LLMs with SMT solvers like Z3 employs a generate-and-check pipeline where candidate solutions are formally verified, and counterexamples are fed back to the model for refinement [48](#). Applying this principle here means creating custom solvers for the failing puzzle types. These solvers will act as a source of truth, providing a definitive standard against which the LLM's reasoning can be measured and corrected.

The second pillar is the enforcement of strict structural consistency. The "format chaos" that plagued the 78K mega-dataset strategy must be explicitly avoided. The CoT traces generated by the ground-truth solvers should adhere to a predefined template or formal grammar. This ensures that every training example has the same syntactic structure, removing any ambiguity and preventing the model from developing brittle dependencies on specific token sequences. This aligns with the concept of Structured CoTs (SCoTs) for code generation, which uses a novel prompting technique to guide the LLM to produce outputs that conform to a desired structure [13](#). Furthermore, evaluation must move beyond final-answer accuracy to assess the reasoning trace itself. Benchmarks like PRISM-Bench and metrics like ChainEval are designed precisely for this purpose, evaluating step-level consistency and logical coherence [15](#) [18](#) [50](#). By verifying each step in the CoT, we can identify and penalize models that arrive at the right answer for the wrong reasons—a flaw that high textual fluency scores might otherwise mask [44](#).

The third and most powerful pillar is the implementation of a neuro-symbolic verification loop. This creates a closed-loop system where the LLM generates a candidate solution, a separate symbolic engine validates it, and feedback is provided to refine future attempts. This approach transforms the training process from a static, one-shot learning event into a dynamic, iterative refinement cycle. When a generated CoT is found to be invalid by the symbolic verifier, instead of simply receiving a negative reward for a wrong final answer, the model receives specific feedback about *which step* was incorrect. This guidance is far more effective for learning. The framework described in [48](#) exemplifies this, where a counterexample produced by the Z3 solver is included in the next prompt to guide the LLM to rule out the specific failing case. This method forces the model to develop a truly

symbolic reasoning capability, as it learns that its output must conform to the formal rules of the domain to be considered valid. This is a direct response to the problem of "post-hoc explanation," where models fabricate rationales [3](#). With a verification loop, fabrication is impossible; the model must generate a trace that the symbolic engine can successfully validate. This methodology is also seen in systems like SymCode, which instructs the LLM to translate a problem into a formal representation manipulated by a Computer Algebra System [23](#), and EduRAG-Compose, which uses a caching layer to store frequently occurring query patterns and intermediate results, improving efficiency and reducing inference latency [49](#).

A Multi-Stage Framework for Implementing Verifiable CoT

Based on the synthesis of the problem's root causes and the principles of neuro-symbolic verification, a multi-stage framework is proposed to systematically replace the flawed baseline with a robust, generalizable reasoning system. This framework is designed to surgically address the issues of memorization, format inconsistency, and lack of generalization identified in previous experiments. It prioritizes the creation of high-fidelity training data and employs a targeted training strategy to minimize disruption to the model's existing capabilities.

Stage 1: Ground-Truth Generation and Validation Set Creation. The initial and most critical phase is to establish a foundation of perfectly consistent, verifiable CoT examples for the "failing_ids." This begins by isolating the failing puzzle instances from the main `train.csv` dataset. For each distinct puzzle type among the failures (e.g., cipher, bit-manipulation, equation_symbolic), a dedicated programmatic solver must be developed. These solvers are not just for finding answers; their primary function is to generate step-by-step reasoning traces that are logically sound and syntactically identical. This process mirrors the data synthesis method used by rStar-Math, which employs extensive MCTS rollouts to create high-quality reasoning trajectories that are then verified by code execution [47](#). The output of this stage is a curated dataset of solved failing puzzles, where each entry consists of the original problem, a ground-truth answer, and a fully validated CoT trace generated by the corresponding solver. This dataset becomes the gold-standard resource for all subsequent training phases.

Stage 2: Curated Hybrid Dataset Construction and Adapter Training. To avoid the pitfalls of previous attempts, such as format mixing and loss of memorization, a new training regimen is required. Instead of attempting to fine-tune the existing 0.86 model—which is deeply entrenched in its memorization-based strategy—a fresh adapter or head should be trained on a carefully constructed hybrid dataset. This dataset should contain two types of records: 1. **Successful Puzzle Records:** All records from the original `train.csv` whose existing CoTs are already consistent with the ground-truth solvers developed in Stage 1. These records reinforce the model's correct handling of non-failing puzzles. 2. **Failing Puzzle Records:** All `failing_ids` records, now completely replaced with their newly generated, perfectly consistent CoTs from Stage 1. Crucially, any record from the original dataset where the CoT is inconsistent, ambiguous, or suspected to be part of the "fake" all-zeros mapping should be excluded. This creates a clean, homogeneous training set that is free from conflicting signals. Training a new adapter on this dataset allows the model to learn the new, verifiable reasoning patterns without being corrupted by the old, faulty ones. This strategy is analogous to the Farl framework, which promotes reasoning-dominant behavior by integrating memory unlearning with reinforcement learning to suppress retrieval-based shortcuts during training [3](#). The goal is to teach the model a new, superior way to reason that it can apply to the failing puzzles, while retaining its ability to handle the rest of the dataset correctly.

Stage 3: Neuro-Symbolic Verification Loop for Iterative Refinement. Once a model trained on the curated dataset is available, its reasoning traces must be put through a rigorous verification process. This stage implements the neuro-symbolic loop discussed previously. For any given puzzle, the model generates a candidate CoT and final answer. This candidate trace is then passed to the appropriate symbolic solver (from Stage 1) for validation. The solver checks the logical correctness of each step according to the formal rules of the puzzle domain. If the entire trace is valid, the answer is accepted. If any step is invalid, a counterexample is generated. This counterexample—the specific point of failure—is then fed back to the model as part of a new training or fine-tuning iteration, with the instruction to "Refine the reasoning to rule out this counterexample." This feedback-driven refinement process, similar to the one described in [48](#), forces the model to develop a truly symbolic understanding of the problem-solving process. It cannot rely on memorization or plausible-sounding text; its output must be demonstrably correct. This iterative process continues until the model can consistently generate valid CoT traces for the formerly failing puzzle types. This approach directly addresses the "faithfulness" problem, ensuring that the generated CoT is an accurate reflection of the model's reasoning process [34](#) [52](#).

Advanced Techniques for Reinforcement and Generalization

While the multi-stage framework provides a solid foundation for improving reasoning on the failing puzzle types, achieving robust generalization and maximizing performance requires advanced techniques, particularly in the domain of reinforcement learning (RL). The initial GRPO RL attempt failed, likely due to a poorly defined reward signal and insufficient exploration of the action space. However, by incorporating insights from recent neuro-symbolic and RL research, a more effective RL-based strategy can be formulated to guide the model towards generating verifiable CoTs.

A refined RL approach must focus on creating a meaningful and informative reward function. The primary limitation of the previous attempt was the lack of a clear signal for the model to follow. A successful reward function should be based on the verifiability of the generated CoT, not just the final answer. As demonstrated by the rStar-Math framework, a pairwise preference model can be trained to evaluate intermediate reasoning steps more effectively than methods using noisy absolute scores [47](#). This can be adapted for RL by defining a reward model that assesses the validity of each step in the CoT. For example, a positive reward could be assigned for each correct step, and a large negative reward could be applied for any step that is refuted by the symbolic verifier. This dense reward signal provides continuous feedback throughout the generation process, guiding the policy model far more effectively than a single binary reward at the end. This strategy moves beyond rewarding only the final answer and incentivizes the model to build a correct reasoning chain from start to finish, directly addressing the "right for the wrong reasons" problem [44](#).

Furthermore, the exploration-exploitation trade-off inherent in RL presents a significant challenge [4](#). The discrete, symbolic nature of CoT generation forces the model into a high-certainty regime, which can lead to premature commitment and hinder exploration of alternative reasoning paths [4](#). To combat this, curriculum learning can be employed as a theoretically grounded method for training the RL agent [4](#). The training process would begin with simpler versions of the failing puzzles, gradually increasing in complexity. This staged approach helps mitigate the distributional mismatch that can cause training to fail, allowing the agent to build its reasoning capabilities incrementally [4](#). Additionally, techniques like self-consistency, which sample multiple reasoning paths and select the most consistent answer, can be integrated into the RL agent's decoding strategy to encourage exploration [54](#) [69](#). By generating several potential CoTs for a given puzzle and selecting the one that is most frequently proposed or most easily verified by

the symbolic engine, the agent can overcome premature commitment and discover more robust solutions.

Finally, to ensure that the skills learned for the failing puzzle types generalize to other, unseen problems, the training process must incorporate mechanisms that promote abstract reasoning. The "forgetting-augmented" aspect of the Farl framework, which integrates memory unlearning with reinforcement learning, is particularly relevant [3](#). By actively suppressing retrieval-based shortcuts during training, the model is forced to rely on its newly acquired reasoning capabilities. This can be implemented by periodically exposing the model to distractor examples that share superficial similarities with the target puzzles but require different reasoning processes. The model must learn to ignore these distractors and apply the correct symbolic logic, thereby strengthening its generalizable reasoning abilities. Experiments with the Farl method showed significant gains in out-of-domain tasks, highlighting the effectiveness of this approach in promoting transferable skills [3](#). By combining a step-level verification reward, curriculum learning, and active forgetting techniques, the RL agent can be guided to not only solve the specific failing puzzles but also to develop a more profound and adaptable symbolic reasoning capability that benefits its overall performance.

Actionable Recommendations and Path Forward

To surpass the 0.86 leaderboard score, a departure from incremental data augmentation is imperative. The evidence overwhelmingly indicates that the current baseline is a fragile equilibrium sustained by memorization, and attempts to modify it without a comprehensive, principled strategy will inevitably fail. The following actionable recommendations outline a clear, phased path forward designed to replace the flawed "fake CoT" mechanism with a robust, verifiable, and generalizable reasoning system.

First and foremost, the immediate priority must be the development of programmatic ground-truth solvers for each of the failing puzzle types identified in the `failing_ids` JSON file. This foundational step, detailed in Stage 1 of the proposed framework, is non-negotiable. These solvers will serve as the ultimate arbiter of correctness, providing a source of perfectly consistent and verifiable CoT examples. Without this objective standard, any attempt to train or verify the model's reasoning is based on subjective and potentially flawed human-generated data. The methodology should mirror that of rStar-Math, focusing on generating high-quality, machine-verifiable reasoning traces that can be checked for both logical and computational accuracy [47](#).

Second, the "All-or-nothing real CoTs" hypothesis (Strategy 14) should be executed with rigor, but with a crucial refinement informed by this analysis. Instead of applying it to the entire dataset, it should be tested on a carefully curated subset. Create a new training dataset composed exclusively of the `failing_ids` records, now paired with their newly generated, solver-verified CoTs. Train a fresh adapter on this dataset while keeping the original 0.86 model frozen. The result of this test will be definitive: if the new adapter performs well on the failing puzzles, it confirms that the model is capable of learning the new format and reasoning style when freed from the corrupting influence of the old, mixed-format data. If it fails, it reinforces the hypothesis that the model is hopelessly dependent on its memorization-based strategy.

Third, the evaluation methodology must evolve. The ultimate goal is not merely to achieve a higher final score but to build a model that reasons correctly. Therefore, the evaluation process must move beyond final-answer accuracy to include a step-level verification metric, similar to the Step F1-Score used in the EngChain benchmark [44](#). For each prediction, the generated CoT must be programmatically checked against the rules of the puzzle domain. This will provide a direct measure of the model's procedural reasoning capability and expose cases where the model gets the right answer for the wrong reasons. This multidimensional evaluation is critical for diagnosing genuine reasoning errors versus mere memorization [19](#) [51](#).

Finally, if the curated dataset training shows promise, the next step should be to implement the neuro-symbolic verification loop. This closed-loop system, inspired by frameworks that integrate LLMs with SMT solvers [48](#), will be the engine for iterative refinement. The model will generate CoTs, receive specific feedback on their logical flaws, and learn to produce increasingly valid and faithful reasoning traces. This process directly confronts the issue of faithfulness and aims to instill a genuine symbolic reasoning capability in the model, rather than teaching it to mimic one [34](#). In parallel, a more sophisticated reinforcement learning strategy should be explored, using a reward function based on step-level correctness verified by the symbolic engine. This dense reward signal, combined with curriculum learning and techniques to promote exploration, offers the best path toward robust generalization and a significant leap in performance. By following this structured, evidence-based approach, it is possible to construct a new reasoning paradigm that not only solves the failing puzzle instances but also establishes a more reliable and defensible foundation for future advancements.

Reference

1. Embracing mathematics identity in an African-centered school <https://search.proquest.com/openview/f326f6b79b8bf48a93702bb01536ddf1/1?pq-origsite=gscholar&cbl=18750>
2. The International Dictionary of Psychoanalysis - PDFCOFFEE.COM <https://pdfcoffee.com/the-international-dictionary-of-psychoanalysis-pdf-free.html>
3. Reasoning or Retrieval? A Study of Answer Attribution on Large ... <https://arxiv.org/html/2509.24156v2>
4. The Theoretical Benefits and Limitations of Latent Chain-of-Thought... <https://openreview.net/forum?id=q7Nhu2Fw11>
5. [PDF] Boosting Multi-modal Keyphrase Prediction with Dynamic Chain-of ... <https://aclanthology.org/2025.emnlp-main.798.pdf>
6. Alleviating confirmation bias in perpetually dynamic environments <https://www.sciencedirect.com/science/article/pii/S0952197624012156>
7. [PDF] Your Prompts Are Not Safe: Output-Free Membership Inference via ... <https://ojs.aaai.org/index.php/AAAI/article/view/40839/44800>
8. Contrastive Learning-Enhanced Chain-of-Thought Optimization for ... <https://ieeexplore.ieee.org/iel8/6287639/11323511/11318557.pdf>
9. Evaluation of DeepSeek-R1 and ChatGPT-4o on the Chinese ... <https://www.nature.com/articles/s41598-025-31874-6>
10. FinChain: A Symbolic Benchmark for Verifiable Chain-of ... - arXiv <https://arxiv.org/html/2506.02515v4>
11. [PDF] SymPyBench: A Dynamic Benchmark for Scientific Reasoning with ... <https://aclanthology.org/2026.eacl-industry.8.pdf>
12. FinChain: A Symbolic Benchmark for Verifiable Chain-of ... - arXiv <https://arxiv.org/html/2506.02515v3>
13. Structured Chain-of-Thought Prompting for Code Generation <https://dl.acm.org/doi/10.1145/3690635>
14. [PDF] A Multi-Agent LLM Framework for Complex Scientific Problem-Solving <https://openreview.net/pdf?id=zT846b7OP5>
15. A Benchmark of Puzzle-Based Visual Tasks with CoT Error Detection <https://arxiv.org/html/2510.23594v1>

16. Puzzle Everything: Benchmarking Knowledge Reasoning through... <https://openreview.net/forum?id=jjrzADWOD5>
17. A Benchmark for Multimodal, Open-Ended Reasoning in Puzzlehunts <https://arxiv.org/html/2506.06211v2>
18. (PDF) PRISM-Bench: A Benchmark of Puzzle-Based Visual Tasks ... https://www.researchgate.net/publication/396967761_PRISM-Bench_A_Benchmark_of_Puzzle-Based_Visual_Tasks_with_CoT_Error_Detection
19. Multi-Dimensional Evaluation of Auto-Generated Chain-of-Thought ... <https://www.mdpi.com/2673-2688/7/1/35>
20. [PDF] Evaluating Step-by-step Reasoning Traces: A Survey - ACL Anthology <https://aclanthology.org/2025.findings-emnlp.94.pdf>
21. 机器学习2026_2_5 - arXiv每日学术速递 <https://www.arxivdaily.com/thread/76282>
22. [PDF] The 14th International Joint Conference on Natural Language ... <https://aclanthology.org/2025.ijcnlp-srw.pdf>
23. SymCode: A Neurosymbolic Approach to Mathematical Reasoning ... <https://arxiv.org/html/2510.25975v2>
24. [PDF] Efficient Rectification of Neuro-Symbolic Reasoning Inconsistencies ... <http://www.lamda.nju.edu.cn/publication/hu25aaai.pdf>
25. Learning where and when to reason in neuro-symbolic inference <https://openreview.net/forum?id=en9V5F8PR->
26. MILE: Memory-Interactive Learning Engine for Neuro-Symbolic ... <https://ieeexplore.ieee.org/iel8/6287639/10380310/10680497.pdf>
27. The underlying neural bases of the reversal error while solving ... <https://pmc.ncbi.nlm.nih.gov/articles/PMC9755259/>
28. Symbolic equation solving via reinforcement learning - ScienceDirect <https://www.sciencedirect.com/science/article/pii/S0925231224015030>
29. What is chain of thought (CoT) prompting? - IBM <https://www.ibm.com/think/topics/chain-of-thoughts>
30. Testing Explainability of Chain of Thought for Large Language Models <https://www.mdpi.com/2076-3417/16/7/3112>
31. FinChain: A Symbolic Benchmark for Verifiable Chain-of-Thought ... https://www.researchgate.net/publication/392372534_FinChain_A_Symbolic_Benchmark_for_Verifiable_Chain-of-Thought_Financial_Reasoning
32. Download book PDF - Springer Nature <https://link.springer.com/content/pdf/10.1007/978-1-4615-8157-4.pdf>

33. [PDF] When Chain of Thought is Necessary, Language Models Struggle to ... <https://arxiv.org/pdf/2507.05246>
34. Arxiv今日论文| 2026-02-25 - 闲记算法 http://lonepatient.top/2026/02/25/arxiv_papers_2026-02-25
35. The 2025 Conference on Empirical Methods in Natural Language ... <https://aclanthology.org/events/emnlp-2025/>
36. Language - UNESCO Digital Library <https://unesdoc.unesco.org/ark:/48223/pf0000002379>
37. [PDF] The Totality Theorem: Resolution of the Closed Universe Paradox ... <https://zenodo.org/records/18715922/files/The%20Totality%20Theorem%20v141.pdf?download=1>
38. Sources and Detection of Dark Matter and Dark Energy in the Universe <https://link.springer.com/content/pdf/10.1007/978-3-662-04587-9.pdf>
39. URSA: Understanding and Verifying Chain-of-thought Reasoning in ... <https://arxiv.org/html/2501.04686v4>
40. [PDF] Ascending Mathematical Solution Quality via Analogical-Problem ... <https://aclanthology.org/2025.findings-emnlp.232.pdf>
41. [PDF] VRAgent-R1: Boosting Video Recommendation with MLLM-based ... <https://ojs.aaai.org/index.php/AAAI/article/view/37152/41114>
42. TopoBench: Benchmarking LLMs on Hard Topological Reasoning https://www.researchgate.net/publication/401910745_TopoBench_Benchmarking_LLMs_on_Hard_Topological_Reasoning
43. MeRF: Motivation-enhanced Reinforcement Finetuning for Large ... <https://arxiv.org/html/2506.18485v1>
44. EngChain: A Symbolic Benchmark for Verifiable Multi-Step ... - arXiv <https://arxiv.org/html/2511.01650v1>
45. TIR-Bench: A Comprehensive Benchmark for Agentic Thinking-with ... <https://arxiv.org/html/2511.01833v1>
46. LogicSkills Benchmark Tests LLMs' Reasoning Abilities - LinkedIn https://www.linkedin.com/posts/brian-rabern_logicskills-a-structured-benchmark-for-formal-activity-7426747469096701952-wLb0
47. rStar-Math: Small LLMs Can Master Math Reasoning with Self ... <https://openreview.net/forum?id=5zwF1GizFa>
48. A Hybrid Framework of Reasoning optimised LLMs and SMT Solvers <https://arxiv.org/html/2508.00419v1>
49. Hybrid retrieval generation for structured reasoning with large ... <https://link.springer.com/article/10.1007/s44163-026-01059-9>

50. FinChain: A Symbolic Benchmark for Verifiable Chain-of ... - arXiv <https://arxiv.org/html/2506.02515v2>
51. Measuring and Improving Chain-of-Thought Reasoning in Vision ... https://www.researchgate.net/publication/373822370_Measuring_and_Improving_Chain-of-Thought_Reasoning_in_Vision-Language_Models
52. [PDF] Measuring and Improving Faithfulness of Chain-of-Thought Reasoning <https://aclanthology.org/2024.findings-emnlp.882.pdf>
53. Quantifying and Optimizing Human-Model Alignment in Chain-of ... <https://arxiv.org/html/2511.06168v1>
54. Chain-of-Thought Reasoning Without Prompting - ResearchGate https://www.researchgate.net/publication/397198105_Chain-of-Thought_Reasoning_Without_Prompting
55. Measuring and Improving Chain-of-Thought Reasoning in Vision ... <https://arxiv.org/html/2309.04461v2>
56. 1 Introduction - arXiv <https://arxiv.org/html/2604.17140v1>
57. Computer Science Feb 2023 - arXiv <http://arxiv.org/list/cs/2023-02?skip=2425&show=2000>
58. Computer Science Feb 2023 - arXiv <http://arxiv.org/list/cs/2023-02?skip=1740&show=2000>
59. GSM-Symbolic: Understanding the Limitations of Mathematical... <https://openreview.net/forum?id=AjXkRZlvjB>
60. Chain-of-thought prompting elicits reasoning in large language ... <https://dl.acm.org/doi/10.5555/3600270.3602070>
61. [PDF] The Effects of Chain-of-Thought Prompting on the Performance of ... https://terra-docs.s3.us-east-2.amazonaws.com/IJHSR/Articles/volume8-issue7/IJHSR_2026_87_20.pdf
62. Cyberspace Safety and Security - Springer Nature <https://link.springer.com/content/pdf/10.1007/978-3-030-73671-2.pdf>
63. Arxiv今日论文| 2026-05-12 - 闲记算法 http://lonepatient.top/2026/05/12/arxiv_papers_2026-05-12.html
64. Computer Science - arXiv <https://www.arxiv.org/list/cs/new?skip=25&show=1000>
65. MM-CoT: A Benchmark for Probing Visual Chain-of-Thought ... - arXiv <https://arxiv.org/html/2512.08228v1>
66. [PDF] CoMAT: Chain of Mathematically Annotated Thought Improves ... <https://aclanthology.org/2025.emnlp-main.1024.pdf>
67. FormalMATH: Benchmarking Formal Mathematical Reasoning of ... <https://openreview.net/forum?id=mGzeNcMGFu>

68. [PDF] FinChain: A Symbolic Benchmark for Verifiable Chain-of ... - arXiv <https://arxiv.org/pdf/2506.02515>
69. [PDF] SELF-CONSISTENCY IMPROVES CHAIN OF THOUGHT ... <http://webdocs.cs.ualberta.ca/~dale/papers/iclr23b.pdf>
70. Measuring AI Reasoning: A Guide for Researchers - arXiv <https://arxiv.org/html/2605.02442v1>
71. Evaluation of LLMs for mathematical problem solving - ScienceDirect <https://www.sciencedirect.com/science/article/pii/S305047592600401X>
72. The Structure of Economics_ A Mathematical Analysis - Academia.edu https://www.academia.edu/103956845/The_Structure_of_Economics_A_Mathematical_Analysis